Dataset Details

AI-Powered Telecom Network Operations Assistant
Dataset documentation (submission deliverable 6)

Item	Value
Dataset name	5G Network Operations dataset (5G_netops_data_100K)
Source	Hugging Face Hub, dataset `fenar/5g_netops`
Link	https://huggingface.co/datasets/fenar/5g_netops
Format	CSV compressed with xz (5G_netops_data_100K.csv.xz), 100,000 rows × 27 columns
Licence	Apache-2.0 (as published on the dataset page)
Pinned version	revision 75f94e94097e..., SHA-256 006b685455531c41...

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1. Dataset name, source and link

Field	Value
Dataset name	5G Network Operations dataset — file 5G_netops_data_100K.csv.xz
Dataset source	Hugging Face Hub, publisher fenar (public dataset)
Dataset link	https://huggingface.co/datasets/fenar/5g_netops
Exact version used	commit `75f94e94097eb72efcd95ef417d4be171ac31bb6`
Integrity check	SHA-256 `006b685455531c414fcb4872aa8e10ca205091c41a0cf7395f276b6d86c96e98` — verified on every download
How it is obtained	`python -m src.preprocessing.download_dataset` downloads the pinned file to `data/raw/`; the raw file is never modified
Sample in the repository	`data/sample_data/5G_netops_sample_500.csv` — the first 500 rows, byte-for-byte as in the dataset

2. Data format

Property	Value
File type	Comma-separated values (CSV), header row, UTF-8, xz-compressed (6.3 MB; 44.8 MB in memory)
Rows × columns	100,000 × 27
Missing values	0 in every column
Duplicates	0 exact duplicate rows
Unit of observation	one row = one network-performance observation for a cell/site: KPIs, alarms, configuration, security counters, weather context and a location
Identifiers	none in the source; the project adds record_id ROW-000001 … ROW-100000 in file order
Time information	season only — no timestamps, so no time-series analysis is possible

Three real raw rows (selected columns):

Season	Latency (ms)	Packet Loss Rate (%)	Signal Strength (dBm)	Weather	City, State	Fault Occurrence Rate (%)
Winter	74.336	6.438	-110.12	Clouds	Schneiderville, MN	33.5824
Fall	90.651	5.18	-77.96	Clear	Lake Heidiside, IN	31.1710
Winter	78.068	8.688	-114.57	Clouds	Howardview, KS	16.3431

3. Key fields

All 27 columns. 'Name in project' is the canonical name used in the code (`src/config/schema.yaml`); statistics are computed on all 100,000 rows.

Column in dataset	Name in project	Type	Role	Min	Median	Max
Season	`season`	category	context	–	–	–
Cell Availability (%)	`cell_availability_pct`	float	kpi	74.342	89.401	99
MTTR (hours)	`mttr_hours`	float	operational	0.5	5.25	10
Throughput (Mbps)	`throughput_mbps`	float	kpi	114.992	549.212	1283.86
Latency (ms)	`latency_ms`	float	kpi	4.618	77.216	175.844
Packet Loss Rate (%)	`packet_loss_pct`	float	kpi	0.1	6.24	17
Call Drop Rate (%)	`call_drop_pct`	float	kpi	0	0	6.9003
Handover Success Rate (%)	`handover_success_pct`	float	kpi	90	95.02	100
Alarm Count	`alarm_count`	int	operational	0	10	26
Critical Alarm Count	`critical_alarm_count`	float	operational	0	6.7453	13.8642
Parameter Changes	`parameter_changes`	int	operational	0	10	19
Successful Configuration Changes (%)	`config_success_pct`	float	operational	90	94.49	99
Data Usage (GB)	`data_usage_gb`	float	load	10	25.16	50
User Count	`user_count`	int	load	100	5060	9999
Signal Strength (dBm)	`signal_strength_dbm`	float	kpi	-120	-90.06	-60
Jitter (ms)	`jitter_ms`	float	kpi	0.1	25	50
Connection Setup Success Rate (%)	`setup_success_pct`	float	kpi	90	94.96	99
Security Incidents	`security_incidents`	int	security	1	10	19
Authentication Failures	`auth_failures`	int	security	0	8	63
Temperature (°C)	`temperature_c`	float	environment	-10	17.47	40
Humidity (%)	`humidity_pct`	float	environment	20	54.9	90
Weather	`weather`	category	environment	–	–	–
Issue Reported	`issue_reported`	bool	flag	–	–	–
City	`city`	text	location	–	–	–
State	`state`	category	location	–	–	–
Zip	`zip`	zip	location	–	–	–
Fault Occurrence Rate (%)	`fault_rate_pct`	float	target	0	39.4552	81.721

Categorical fields: Season (Winter, Spring, Summer, Fall), Weather (Clear, Clouds, Rain, Snow), Issue Reported (yes/no), State (59 US states and territories), City and Zip (random locations).

4. Dataset purpose in this project

- **Evidence base for retrieval-augmented analysis.** Each of the 100,000 records is a historical observation the assistant can retrieve as evidence when an engineer describes a problem.
- **Training data for the ML models.** Fault occurrence risk (regression on Fault Occurrence Rate), throughput (regression) and a category classifier.
- **Source of truth.** Every number shown to a user is either a raw value from this file or computed from it by a documented rule; the LLM is never allowed to introduce values that are not in the evidence.

> **Limits of the data**
> The dataset has no incident labels, free text, timestamps or cell identifiers. Categories, severity and priority are therefore derived by documented rules and labelled as derived. Phase 1 analysis shows the data is synthetic (uniform distributions, random locations, fault rate almost a linear formula of seven columns), so root causes are reported as hypotheses.

5. Data preprocessing performed

Run with `python -m src.preprocessing.build_all` (steps 1–7) — rows in 100,000, rows out 100,000, rows dropped 0.

Step	What is done	Result
1. Download	pinned revision, SHA-256 verification	raw file in data/raw/, never modified
2. Validate (raw)	types, ranges (e.g. signal −140…−44 dBm, % within 0–100), allowed categories, missing values, whitespace	0 errors in 100,000 rows
3. Clean	snake_case names, typed values (exact float parsing), yes/no → boolean, ZIP as 5-character text, exact duplicates dropped, no imputation	data/processed/records.parquet
4. Quality flags	suspicious but valid patterns are flagged, never changed	dq_critical_exceeds_total_alarms 15,749, dq_availability_at_cap 11,872, dq_setup_success_at_cap 10,022, dq_data_usage_at_floor 19,682
5. Record IDs	stable ID per row in file order	ROW-000001 … ROW-100000
6. Derived features	18 conditions from worst-decile thresholds (signal: −110 dBm domain threshold); rule-based category, points-based severity, priority	data/processed/incidents.parquet (62 columns)
7. Chunking	one chunk per record: search_text (full, for BM25) and embedding_text (qualitative, for embeddings), built only from the record's own fields	27,117 distinct embedding texts
8. Indexing	all-MiniLM-L6-v2 embeddings into ChromaDB; BM25 sparse matrix	data/processed/chroma/, data/processed/bm25/

A test compares every processed value with the raw CSV text, proving preprocessing neither alters nor fabricates a measurement.

Derived conditions (first 8 of 18):

Condition	Metric	Rule	Method
Low cell availability	cell_availability_pct	< 79.4339	percentile
Low throughput	throughput_mbps	< 304.248	percentile
High latency	latency_ms	> 129.922	percentile
High packet loss	packet_loss_pct	> 11.88	percentile
High call drop rate	call_drop_pct	> 3.1338	percentile
Low handover success	handover_success_pct	< 91.01	percentile
Weak signal strength	signal_strength_dbm	< -110	domain
High jitter	jitter_ms	> 44.98	percentile

Example chunk (embedding_text) for ROW-000001:

```
Derived conditions: weak signal strength (weak_signal), high data usage (high_data_usage), many parameter
    changes (high_parameter_changes). Derived category: Radio / Signal Quality (secondary: Congestion /
    Load, Configuration). Derived severity: LOW. Derived priority: P4. Weather Clouds, season Winter.
    Customer issue reported: yes.
```